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Last updated: August 28, 2026

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Education	2012-2018	Ph.D. (Physics) Homi Bhabha National Institute (HBNI), Mumbai-400094, India. CI: Bhabha Atomic Research Centre (BARC) , Mumbai-400085, India. Thesis title: Study of Materials Behavior Under Dynamic Shock Loading Using High Power Lasers Thesis Supervisor: Prof. M. N. Deo, Bhabha Atomic Research Centre, Mumbai, India.
	2009-2011	Master of Science (M.Sc.) in Physics with specialization in Electronics, University of Lucknow, Lucknow, India.
	2006-2009	Bachelor of Science (B.Sc.) , University of Lucknow, Lucknow India.
Teaching & Mentoring Experiences	2023-2025	Assistant Professor (Physics) , Pimpri Chinchwad University, Pune-412106, India. Teaches on topics: Condensed matter physics, High power lasers and their applications, Basics of spectroscopy, Quantum Physics, Magnetism, Nanoscience and Nanotechnology.
	2023-2024	Delivered lectures on Research Ethics and Publications to PhD students
	2019-2022	Co-mentored the undergraduate students in their Research projects
	2016-2017	Co-mentored Master of science student for his summer projects
Research Experience	2025-Current	Postdoctoral Employee, Department of Physics, University of Alabama at Birmingham, Alabama-35205, USA
	2018-2022	Postdoctoral Fellow, Hopkins Extreme Material Institute and Department of Earth & Planetary sciences, Johns Hopkins University, Baltimore-21218, USA.
	Marh-June 2018	Short term project High Pressure & Sychrotron Radiation Physics Division, BARC, Mumbai-400085, India.
Awards and Grants	2012-2017	Department of Atomic Energy, Government of India, Doctoral Research Fellowship.
	2017	Department of Science & Technology (DST-SERB ITS) International Travel Grant.
	2016	Strong Field Science International Travel Grant.
Submitted proposals	2025-2026	Melissa Sims, Vinay Rastogi , Raymond F. Smith et al., Shock Response of BCC-Structured High Entropy Alloys: From Disorder to Discovery, LCLS, Matter in Extreme conditions, 2026 (<i>submitted</i>).
	2024-2025	Vinay Rastogi , ANRF PM-ECRG 2024, “Development of advance analytical techniques and predictive model for shock-compressed Metals (Al, Fe etc.), Ceramics (SiC) and Alloys (Titanium alloys) using Molecular Dynamics Simulations”, Reference No.: 502024009414.

Project Experience	*Dynamic shear strength measurements in alloys *Exploring the influence of high specific grain boundary area on total free energy of alloys. *Kinetics of Phase transition and structural transformation in materials under dynamic compression. *Phase stability of Fe-Si alloy at Super-Earth core conditions. *Development of slurry targets for high frequency (<1Hz) XFEL experiments. *Design and development of a Time Resolved Raman Spectroscopy (TRRS) facility for the dynamically compressed materials and studies on polymeric samples. *Development of Thomson parabola spectrometer (TPS) for laser plasma ion spectrum studies. *Experiments with low density foam targets.
Collaborative Research Work	*Research collaboration with Army Research Directorate, DEVCOM Army Research Laboratory, APG, MD 21005, USA for studying the influence of high specific grain boundary (GB) area can significantly influence the total free energy of a material under extreme conditions. *Collaborative research on structural transformation of materials under dynamic compression with faculties of Johns Hopkins University and University of Alabama at Birmingham, USA. *Research collaboration for Laser matter interaction studies with School of Mathematics and Physics, Queen's University, Belfast, UK. *Performed experiments at Matter at Extreme Condition (MEC end station) at SLAC National facility at Stanford University at USA to study the kinetics of Phase transition and structural transformation in solid state materials under dynamic compression in collaboration with Scientist of Lawrence Livermore National Laboratory, Sandia National Laboratory and Researchers of Princeton University, USA. *Performed experiments at Laboratory for Laser Energetics (LLE) at University of Rochester at USA to study the phase diagram of Fe-Si alloys under ramp compression in collaboration with the Scientists of Lawrence Livermore National Laboratory, USA. *Research collaboration for microstructural and defect studies under extreme conditions with department of physics, Chandigarh University, Lucknow, India.
Membership of Professional Bodies	*Member of European High-Pressure Research Group (EHPRG) *Life Member of Indian Physics Association (IPA) *Member of Indian Nuclear Youth Society (IYNS)
Administrative Responsibilities	*Physics Lab in-charge at Pimpri Chinchwad University *Member of Institution Innovation council (IIC) at Pimpri Chinchwad University *Sports coordinator at Pimpri Chinchwad University
Certificate Courses	*Lean Into Future Teaching, UAB center of Teaching & Learning and UAB Graduate School. *Introduction to Artificial Intelligence (AI) by IBM, Coursera *AI for Scientific Research (Introduction to Data Science and scikit-learn in Python) by Learn Quest, Coursera *Machine Learning Models in Science by Learn Quest, Coursera
Service and Other Activities	Co-Editor (2026): Pressure-Driven Phenomena in Condensed Matter Systems, Frontiers in Physics Journal reviewer: AIP Advances, International Journal of Materials Research, Rapid Communications in Mass Spectrometry, Journal of Energy storage 2025: Organized one day Robotics Industry Workshop for undergraduate students at Autofina Robotics & Automation Pvt. Ltd. India. 2024: Organized one day mentoring session on Accelerators/Incubation- Opportunities in AI for Students and Faculties at Pimpri Chinchwad University, India. 2016: Organizing committee member of CoMPEC-2016 conference at BARC, Mumbai. 2015: Participated in the organization of the visit of the peer team appointed by NAAC. 2012-2013: Training on advanced technologies which includes Lasers, Accelerators, Plasma physics, Ultra-high vacuum systems and Radiation physics.

Publications

1. **Vinay Rastogi**, R. F. Smith, T. S. Duffy, J. K. Wicks, “Phase transition studies of Fe-Si alloys under ramp compression” *JGR: Planet (Accepted)*.
2. M. Kolo, P. Mishra, Y. Singh, R. Parihar, **V. Rastogi** et al. “Synergistic B-Site Modulation in LaCoO₃-La(Cr,Ni)O₃ Hierarchical Nanocomposites: Ultralow Overpotentials and Record Tafel Slopes for Bifunctional Alkaline Water Splitting” *Int. J. Hydrogen Energy (submitted)*.
3. Sunil Rawat, **Vinay Rastogi**, “Role of crystal orientations and loading conditions on the microstructure evolution and void evolution dynamics in single crystal iron: An atomistic investigation” *Mechanics of Materials*, 207 (2025) 105387, <https://doi.org/10.1016/j.mechmat.2025.105387>.
4. **Vinay Rastogi**, R. F. Smith, M. Sims and J. K. Wicks, “Application of ultrafast x-ray lasers in studying the material structure under shock compression” *J. Appl. Phys.*, 137 (2025) 070702, <https://doi.org/10.1063/5.0239330>.
5. **Vinay Rastogi**, S. Chaurasia, “Advances in and Future Perspectives of High-Power Ceramic Lasers” *Photonics*, 11(10), (2024) 942, <https://doi.org/10.3390/photonics11100942>.
6. **Vinay Rastogi**, R. F. Smith, D. C. Swift, R. Briggs, M. G. Gorman, C. Krill, A. L. Coleman, D. E. Fratanduono, J. H. Eggert, C. A. Bolme, F. Coppari, A. Gleason, H. J. Lee, P. Heimann, T. S. Duffy, J. K. Wicks, “Femtosecond diffraction studies of the Sodium Chloride phase diagram under laser-shock compression” *Journal of Applied Physics*, 132, (2022) 85901 <https://doi.org/10.1063/5.0094894>.
7. R. F. Smith, **Vinay Rastogi**, A. E. Lazicki, M. G. Gorman, D. E. Fratanduono, D. C. Swift, C. A. Bolme, A. E. Gleason, F. Coppari, H. J. Lee, E. Granados, P. Heimann, R. G. Kraus, R. E. Rudd, T. S. Duffy, J. H. Eggert, J. K. Wicks, “Development of slurry targets for high repetition-rate XFEL experiments” *Journal of Applied Physics*, 131, (2022) 245901 <https://doi.org/10.1063/5.0095654>.
8. S. Chaurasia, **Vinay Rastogi**, Usha Rao, M. N. Deo, “Development of in situ time-resolved Raman Spectroscopy facility for dynamic shock loading in materials” *Journal of Instrumentation*, 12, (2017) P11008 <https://doi.org/10.1088/1748-0221/12/11/P11008>.
9. **Vinay Rastogi**, S. Chaurasia, U. Rao, C. D. Sijoy, H. K. Poswal, V. Mishra, M. Kumar, M. N. Deo, “In-situ Raman spectroscopic studies of poly-vinyl-toluene under laser-driven shock compression and comparison with hydrostatic experiments” *Journal of Raman Spectroscopy*, 48, (2017) 1300-1306 <https://doi.org/10.1002/jrs.5215>.
10. **Vinay Rastogi**, Usha Rao, S. Chaurasia, C. D. Sijoy, V. Mishra, S. Chaturvedi, M. N. Deo, “Time resolved Raman spectroscopy of polytetrafluoroethylene under laser shock compression” *Appl. Spec.* 71, (2017) 2643-2652 <https://doi.org/10.1177/0003702817726542>.
11. **Vinay Rastogi**, S. Chaurasia, U. Rao, C. D. Sijoy, V. Mishra, M. Kumar, S. Chaturvedi, M. N. Deo, “Time-resolved Raman spectroscopy of polystyrene under laser driven shock compression” *Journal of Raman Spectroscopy* 48 (2017) 1007-1012 <https://doi.org/10.1002/jrs.5166>.
12. **Vinay Rastogi**, S. Chaurasia, D. S. Munda, “Laser induced damage studies in borosilicate glass using nanosecond and sub nanosecond pulses”, *Journal of Non-Crystalline Solids* 463 (2017) 138–147 <https://article/abs/pii/S0022309317301102>.
13. **Vinay Rastogi**, S. Chaurasia, U. Rao, C. D. Sijoy, V. Mishra, M. Kumar, M. N. Deo, S. Chaturvedi, S. M. Sharma, “Raman spectroscopy of laser shocked polystyrene” *Journal of Raman Spectroscopy* 48 (2017) 458-464 <https://doi.org/10.1002/jrs.5046>.
14. S. Chaurasia, C. Kaur, **V. Rastogi**, A. K. Poswal, D. S. Munda, R. K. Bhatia, V. Nataraju, “In situ measurement of ions parameters of laser produced ion source using high resolution Thomson parabola spectrometer” *Journal of Instrumentation* 11 (2016) P08004, <https://article/10.1088/1748-0221/11/08/P08004>

Papers in Prep

- ***Vinay Rastogi**, Melissa Sims, “Shock-induced BCC Polymorphism in Re-based Refractory High Entropy Alloys”.
- *Anirudh Hari, **Vinay Rastogi**, Rohit Hari, Raymond F Smith et al. “Metastability of alpha-corundum phase in shock-compressed single crystalline sapphire”.

Book Chapters

- Vinay Rastogi**, S. Chaurasia, “Laser-induced breakdown mechanisms and damage threshold investigations in optical materials”, Springer, eBook ISBN 978-3-031-64542-6, Print ISBN 978-3-031-64541-9.

**Invited
Lectures**

2021: Study of materials behavior under extreme conditions using dynamic compression, delivered at International Conference on Diverse Emerging Materials and their Applications (ICDEMA-2021), Lucknow (India).

**Oral
Presentations**

2021: Oral Presentation on "Phase diagram studies of the Sodium Chloride under laser driven shock compression" delivered at 10th Asian Conference on High Pressure Research (ACHPR-10), South Korea (Virtual).

2016: Oral Presentation on "Vibrational Spectroscopy of aromatic scintillating polymer under laser-driven shock compression" delivered at 6th International Conference on Perspectives in Vibrational Spectroscopy, Lucknow (India).

2016: Oral Presentation on "Vibrational spectroscopy of polyvinyl-toluene under laser-driven shock compression" delivered at 34th European Conference on Laser Interaction with Matter, Moscow (Russia).

2015: Oral Presentation "Spectroscopic studies of polytetrafluoroethylene under laser driven shock compression" delivered at International Conference on Photons: Multiple & Creative Solutions to Challenges (ICPMCSC), Mumbai (India).

**Conference/
Seminars/
Webinars**

*Raymond Smith, **Vinay Rastogi**, Amy Jenei et al., "Shock study of slurry targets (crystalline grains suspending within an epoxy matrix) for future high repetition rate XFEL diffraction experiments", Bulletin of the American Physical Society 68, 2023.

***Vinay Rastogi**, Raymond F. Smith, Damian C. Swift et al., "Structural transformation determination in Sodium Chloride during shock compression and release" Bulletin of the American Physical Society 67, 2022.

***Vinay Rastogi**, R. F. Smith, D. C. Swift et al., "Phase diagram studies of the Sodium Chloride under laser driven shock compression" 10th Asian Conference on High Pressure Research, Virtual S. Korea, Nov. 21-25, 2021.

*R. Briggs, A. Coleman, S. Zhang, D. McGonegle, F. Coppari, M. Gorman, M. Marshall, R. Smith, O. Ball, R. McWilliams, V. Prakapenka, C. Krill, **Vinay Rastogi** et al., "Probing liquid-liquid phase transitions under dynamic compression: an X-ray diffraction and ab initio MD study of selenium" Bulletin of the American Physical Society 65, 2020.

***Vinay Rastogi**, Raymond F. Smith, Richard J. Briggs et al., "Femtosecond diffraction studies of phase transitions in sodium chloride under dynamic shock compression", COMPRES annual meeting, Montana, USA, Aug. 2- 5 2019.

***Vinay Rastogi**, S. Chaurasia, U. Rao et al., "Vibrational Spectroscopy of Laser Shocked PMMA" 9th High Pressure Mineral Physics Seminar (HPMPS-9) Saint Malo (France) Sept. 24-28, 2017.

***Vinay Rastogi**, Usha Rao, S. Chaurasia et al., "High pressure Raman study polyvinyl-toluene (PVT)" 54th European High Pressure Research Group (EHPRG) Meeting, Bayreuth (Germany), Sept. 4-9, 2016.

***Vinay Rastogi**, Usha Rao, S. Chaurasia et al., "Pump-probe based Raman spectroscopic studies of PTFE under laser driven shock compression" AIP Conf. Proc. 1731, 060025 2016.

***V. Rastogi**, U. Rao, S. Chaurasia et al., "Vibrational spectroscopy of polyvinyl-toluene under laser- driven shock compression" 34th European Conference on Laser Interaction with Matter (ECLIM-2016), Moscow (Russia), Sept. 18-23, 2016.

*U. Rao, **V. Rastogi**, S. Chaurasia et al., "Pump-probe based vibrational spectroscopy of carbon tetrachloride under laser driven shock compression" 34th European Conference on Laser Interaction with Matter (ECLIM-2016), Moscow (Russia), Sept. 18-23, 2016.

***Vinay Rastogi**, S. Chaurasia, U. Rao et al., "Vibrational spectroscopy of aromatic scintillating polymer under laser-driven shock compression" 6th International conference on perspectives in vibrational spectroscopy, Lucknow (India), Nov. 5-8, 2016.

***Vinay Rastogi**, U. Rao, S. Chaurasia et al., "Laser shocked high pressure and time resolved Raman spectroscopy of Polystyrene" DAE-BRNS Symposium on Condensed Matter Physics under Extreme Conditions (CoMPEC 2016), Mumbai (India), April 13-16, 2016.

*U. Rao, S. Chaurasia, C. D. Sijoy, **Vinay Rastogi et al.**, "The effect of geometric confinement on laser driven shock propagation in aluminium-polytetrafluoroethylene (PTFE) layered target" DAE-BRNS

Symposium on Condensed Matter Physics under Extreme Conditions (CoMPEC 2016), Mumbai (India), April 13-16, 2016.

***Vinay Rastogi**, Usha Rao, S. Chaurasia et al., “Spectroscopic studies of polytetrafluoro -ethylene under laser driven shock compression” International conference on Photons: Multiple and creative solutions to challenges (ICPMCSC) Ratnam College, Mumbai (India), Dec. 4-5, 2015.

*Usha Rao, C. D. Sijoy, S. Chaurasia, **Vinay Rastogi et al.**, “Numerical Simulation study of laser driven shock waves in direct and confinement geometry targets” International conference on Photons: Multiple and creative solutions to challenges (ICPMCSC) Ratnam College, Mumbai (India), Dec. 4-5, 2015.

***Vinay Rastogi**, Channpriti Kaur, P. Leshma et al., “Study of laser induced damage mechanism on the surface of BK7 glass using nanosecond and sub nanosecond pulses” 4th International conference on Current development in Atomic, Molecular, Optical & Nanophysics with applications, University of Delhi, Delhi (2015).

*Channpriti Kaur, S. Chaurasia, **V. Rastogi** et al., “K-shell X-ray spectroscopy of laser produced Aluminum plasma” National Laser Symposium (NLS-24), RRCAT, Indore (India), Dec. 2-5, 2015, ISBN: 978-81-903321-6-3 Article: CP-7.21.

*S. Chaurasia, **Vinay Rastogi**, R. K. Bhatia, V. Nataraju, D. S. Munda, S. M. Sharma, “Development of a high resolution and high dispersion Thomson parabola along with time-of-flight detector for laser-plasma produced ions” National Laser Symposium 22, MIT, Manipal, Karnataka (India), Jan. 08-11, 2014.

*S. Chaurasia, **Vinay Rastogi**, R. K. Bhatia et al., “Studies of ions from laser produced carbon plasma using Thomson parabola spectrometer” PLASMA-2013, KIIT, Bhubneshwar (India), Dec. 3-6, 2013.

Newsletters and Reports

2015: S. Chaurasia, **Vinay Rastogi**, R. K. Bhatia et al., “Thomson parabola: A high-resolution spectrometer” BARC Newsletter, <http://www.barc.gov.in/publications/nl/2015/2015050604.pdf>.

2015: BARC Newsletter, Brief Communication, “Facility for time resolved Raman spectroscopy of materials under shocks” <http://www.barc.gov.in/publications/nl/2015/2015070802.pdf>.

Workshops

2024: AI-assisted Peer Review: Using innovative technologies with human oversight by Enago Academy, USA (online).

2014: SERB School on “High intensity laser plasma interaction: theory and simulation” Indian Institute of Technology (IIT) Delhi, Delhi (India).

List of References

1. Prof. Melissa Sims

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2. Prof. June K. Wicks

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3. Dr. Raymond F. Smith

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4. Prof. M. N. Deo

Vice Chancellor (retired) Vinoba Bhave University, Sindoor, Hazaribagh, Jharkhand 825301, India

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